

KCCITM NOIDA • AKTU

WEEK 1 MODULE

PROGRAMMING PRACTICE

Pattern Programs

Star Patterns • Number Patterns

Language Used: **Java**

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Module 1
language used - Java
Star Pattern

1) Square

Logic : • Run 2 loops $\Rightarrow$ outer for rows (1 to n), inner for columns (1 to n)
• Print "*" everytime in inner loop
• Print newline after each row

Code :

```
int n = 5;
for (int i = 1; i <= n; i++) {
    for (int j = 0; j <= n; j++) {
        system.out.print("* ");
    }
    system.out.println();
}
```

```
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

2) Hollow Square

Logic : • same 2 loop as square
• print "*" only if it is first/last row or column
• otherwise print & space

Code :

```
int n = 5;
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= n; j++) {
        if (i == 1 || i == n || j == 1 || j == n) {
            system.out.print("* ");
        }
        else {
            system.out.print(" ");
        }
    }
    system.out.println();
}
```

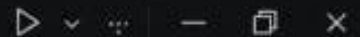
```
* * * * *
*           *
*           *
*           *
*           *
* * * * *
```



First.java x

```
1 public class First {
2     public static void main(String[] args) {
3         // 1. Squares
4         int n = 5;
5         for(int i = 0; i < n; i++) {
6             for(int j = 0; j < n; j++) {
7                 System.out.print("*");
8             }
9             System.out.println();
10        }
11    }
12 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS



Run First x

```
C:\Program Files\Java\jdk-24\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition
2025.1.3\lib\idea_rt.jar=49797" -Dfile.encoding=UTF-8 -classpath C:\collegeTraining\out\production\collegeTraining
First
* * * *
* * * *
* * * *
* * * *
* * * *
```

Process finished with exit code 0

0 0

Ln 10, Col 6 Spaces: 4 UTF-8 CRLF Java: Ready



First.java x



```
1 public class First {
2     public static void main(String[] args) {
3         // 2. Hollow Square
4         int n = 5;
5         for(int i = 0; i < n; i++) {
6             for(int j = 0; j < n; j++) {
7                 if(i == 0 || i == n-1 || j == 0 || j == n-1)
8                     System.out.print("*");
9                 else
10                    System.out.print(" ");
11            }
12            System.out.println();
13        }
14    }
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Run: First x

C:\Program Files\Java\jdk-24\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.1.3\lib\idea_rt.jar=49797" -Dfile.encoding=UTF-8 -classpath C:\collegeTraining\out\production\collegeTraining First

* *
* *
* *

Process finished with exit code 0

Ln 7, Col 52 Spaces: 4 UTF-8 CRLF { } Java Java: Ready

3. Triangle

logic: • outer loop $i = 1$ to n

• inner loop $j = 1$ to i (not n)

• each row print one more star than previous

Code: `int n = 5;`

`for (int i = 1; i <= n; i++) {`

`for (int j = 1; j <= i; j++) {`

`system.out.print("*");`

`}`

`system.out.println();`

`}`

*

* *

* * *

* * * *

* * * * *

4. Inverted Triangle

logic: • no. of stars decrease by 1 after every row

• no. of space increase by 1 after every row to keep it center aligns

Code: `int n = 5;`

`for (int i = n; i >= 1; i--) {`

`for (int j = 0; j < n - i; j++) {`

`system.out.print(" ");`

`}`

`for (int j = 0; j < i; j++) {`

`system.out.print("*");`

`}`

`system.out.println();`

`}`

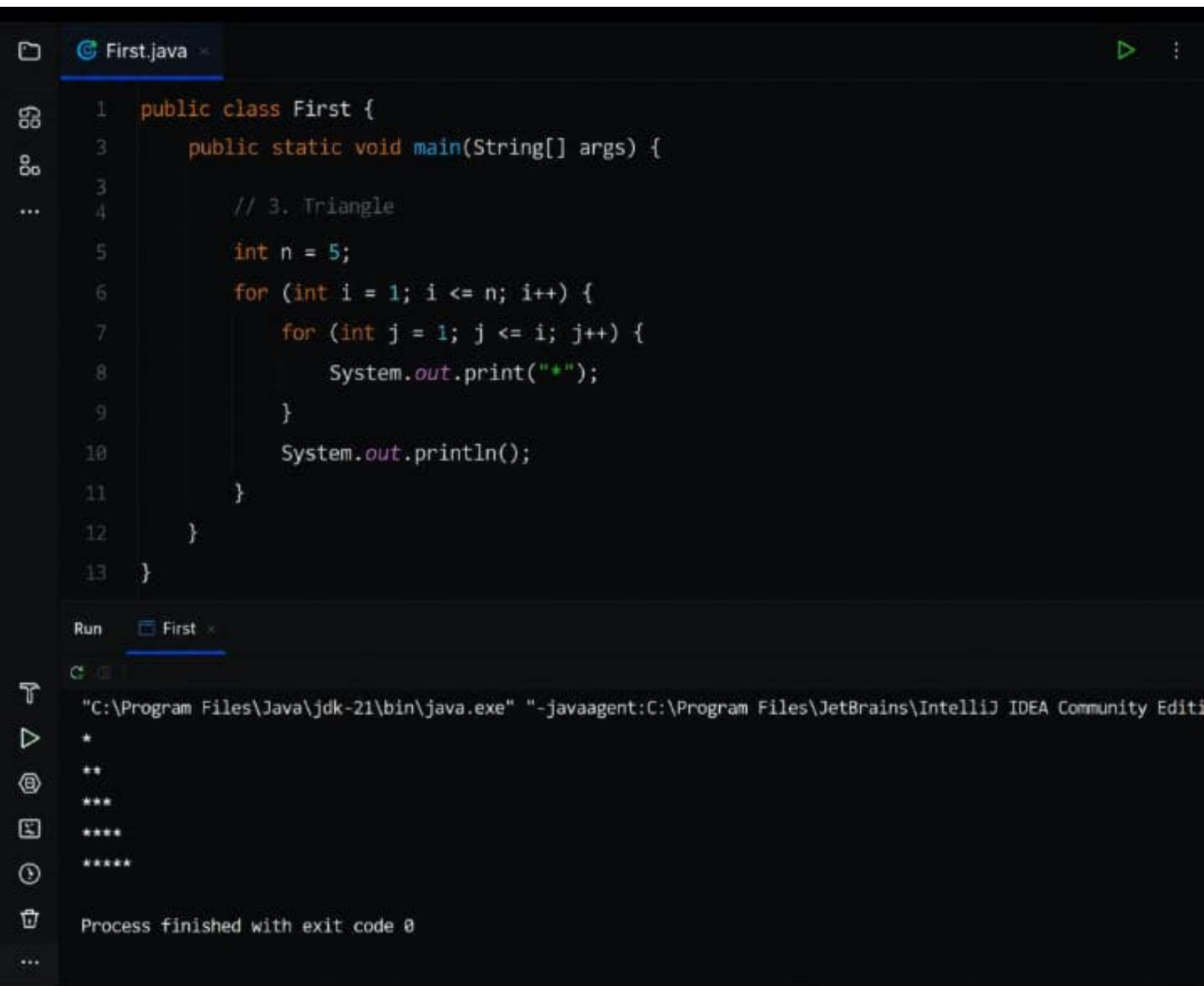
* * * * *

* * * *

* * *

* *

*



First.java x

```
1      public class First {
2          public static void main(String[] args) {
35      //4.    Inverted Triangle
36              int n = 5;
37              for (int i = n; i >= 1 ; i--) {
38                  for (int j = 0; j < n-i; j++) {
39                      System.out.print(" ");
40                  }
41                  for (int j = 0; j < i; j++) {
42                      System.out.print("*");
43                  }
44                  System.out.println();
45              }
46
47      }
```

Run First x

Run First x

"C:\Program Files\Java\jdk-24\bin\java.exe" "-javaagent:C:\P

**

*

Process finished with exit code 0

5. Pyramid

- Logic:
- Row i needs $(n-i)$ space first then $(2i-1)$ stars
 - spaces decrease, star increase as ' i ' increase
 - star always odd count.

Code: `int n = 5;`

```
for (int i = 1; i <= n; i++) {  
    for (int j = 1; j <= n - i; j++) {  
        system.out.print(" ");  
    }  
    for (int j = 1; j <= 2 * i - 1; j++) {  
        system.out.print("*");  
    }  
    system.out.println();  
}
```

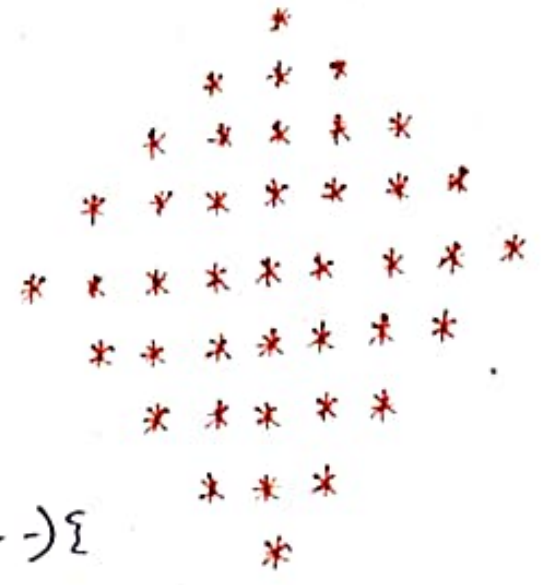


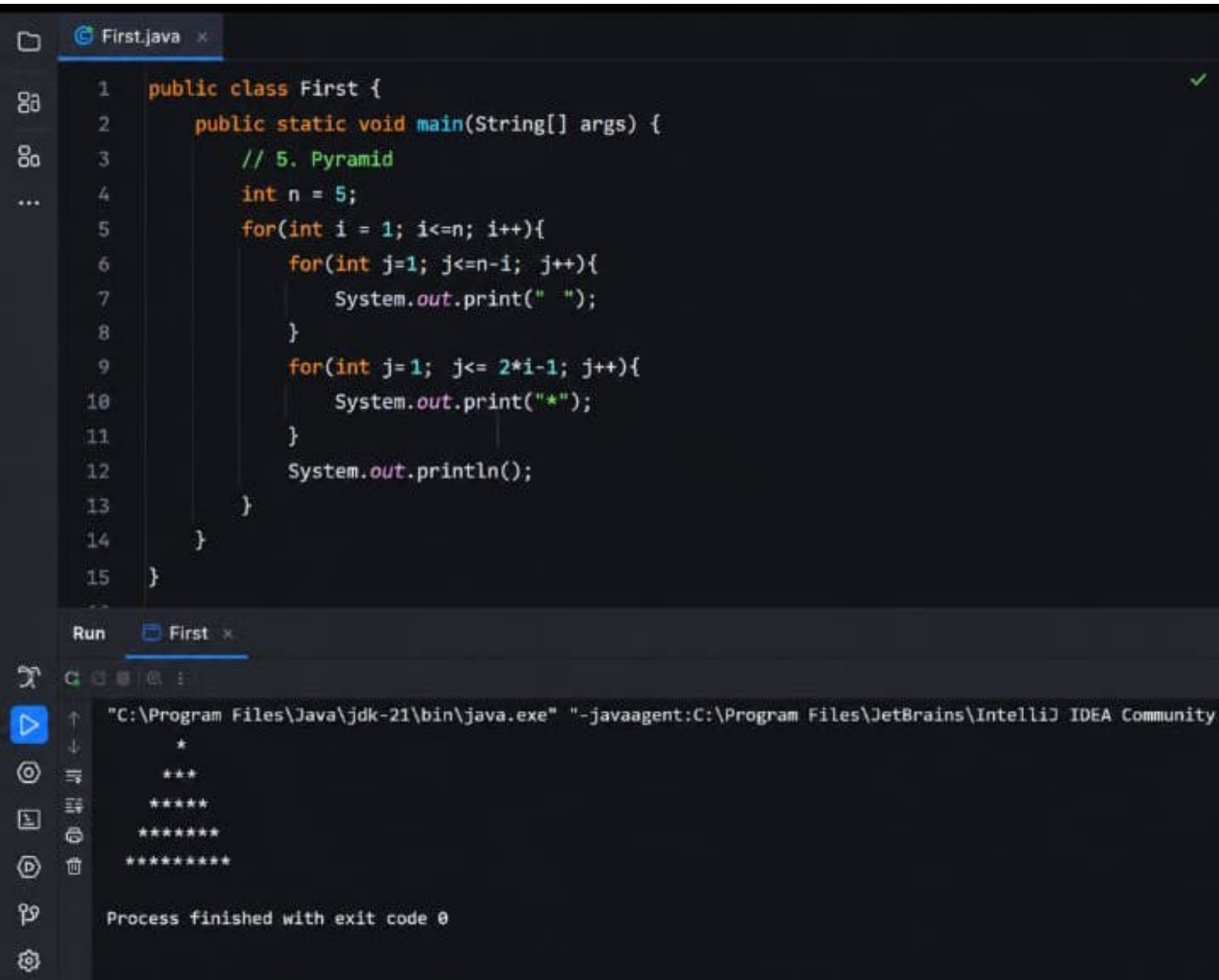
6. Diamond

- Logic:
- upper half = pyramid ($i = 1$ to n)
 - lower half = inverted pyramid ($i = n-1$ down to 1)
 - same formula

Code: `int n = 5;`

```
for (int i = 1; i <= n; i++) {  
    for (int j = 1; j <= n - i; j++) {  
        system.out.print(" ");  
    }  
    for (int j = 1; j <= 2 * i - 1; j++) {  
        system.out.print("*");  
    }  
    system.out.println();  
}  
for (int i = n - 1; i >= 1; i--) {  
    for (int j = 1; j <= n - i; j++) {  
        system.out.print(" ");  
    }  
    for (int j = 1; j <= 2 * i - 1; j++) {  
        system.out.print("*");  
    }  
    system.out.println();  
}
```





First.java

```
3 public class First {
4     public static void main(String[] args) {
5
6         //6. Diamond
7         int n= 5;
8         for(int i =1; i<= n; i++){
9             for(int j=1; j<= n-i;j++){
10                 System.out.print(" ");
11             }for(int j =1; j<= 2*i-1; j++){
12                 System.out.print("**");
13             }System.out.println();
14         }
15         for (int i = n-1; i >=1; i-- ) {
16             for(int j=1; j<= n-i;j++){
17                 System.out.print(" ");
18             }for(int j =1; j<= 2*i-1; j++){
19                 System.out.print("**");
20             }System.out.println();
21         }
22     }
23 }
```

Run First

Run

```
*
***
*****
*****
*****
*****
*****
*****
***
```

7. Half diamond

- logic:
- first half - i goes 1 to n first i stars (increasing)
 - second half - i goes $n-1$ to 1 print i stars (decreasing)
 - No spaces left aligned

code:

```

int n = 5;
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= i; j++) {
        system.out.print("*");
    }
    system.out.println();
}
for (int i = n-1; i >= 1; i--) {
    for (int j = 1; j <= i; j++) {
        system.out.print("*");
    }
    system.out.println();
}

```

```

*
* *
* * *
* * * *
* * * * *
* * * *
* * *
* *
*

```

8. Half diamond inverted

- logic:
- first half i goes n to 1 (decreasing stars)
 - second half i goes 2 to n (increasing stars)

code:

```

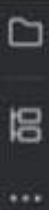
int n = 5;
for (int i = n; i >= 1; i--) {
    for (int j = 1; j <= i; j++) {
        system.out.print("*");
    }
    system.out.println();
}
for (int i = 2; i <= n; i++) {
    for (int j = 1; j <= i; j++) {
        system.out.print("*");
    }
    system.out.println();
}

```

```

* * * * *
* * * *
* * *
* *
*
* *
* * *
* * * *
* * * * *

```



First.java x

```
3 public class First {
4     public static void main(String[] args) {
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32
33
34
35
36
37 //7. Half Diamond
38     int n=5;
39     for (int i = 1; i <= n; i++) {
40         for (int j = 1; j <= i; j++) {
41             System.out.print("*");
42         }
43         System.out.println();
44     }
45     for (int i = n-1; i >= 1; i--) {
46         for (int j = 1; j <= i; j++) {
47             System.out.print("*");
48         }
49         System.out.println();
50     }
51 }
```

Run First x



```
*
**
***
****
*****
****
***
**
*
```

collegeTraining > src > First > main

↑	★★★★★
↓	★★★★★
≡	★★★
≡	★★
≡	★
≡	★★
≡	★★★★
≡	★★★★★
≡	★★★★★

Number Patterns

1. Square

logic: same as star square but print 1 instead of *

code:

```
int n = 5;
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= n; j++) {
        system.out.print("1");
    }
    system.out.println();
}
```

```
1 1 1 1 1
1 1 1 1 1
1 1 1 1 1
1 1 1 1 1
1 1 1 1 1
```

2. Incrementing

logic: • same structure but print j (instead of 1)
• every row print 1 2 3 4 5

code:

```
int n = 5;
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= n; j++) {
        system.out.print(j); // (j)
    }
    system.out.println();
}
```

```
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
1 2 3 4 5
```

3. Right triangle

logic: • outer loop i = 1 to n
• inner loop j = 1 to i
• print j instead of star

code:

```
int n = 5;
for (int i = 1; i <= n; i++) {
    for (int j = 1; j <= i; j++) {
        system.out.print(j);
    }
    system.out.println();
}
```

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

First.java

```
3 public class First {
4     public static void main(String[] args) {
117
118         // Number Pattern
119
120         // 1. Square
121         int n =5;
122         for(int i =1; i <= n; i++){
123             for (int j = 1; j <= n; j++) {
124                 System.out.print("1");
125             }
126             System.out.println();
127         }
128
129     }
```

Run First



"C:\Program Files\Java\jdk-24\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA

```
11111
11111
11111
11111
11111
```

Process finished with exit code 0

First.java

```

1 public class First {
2     public static void main(String[] args) {
3
4         //2. Incrementing
5
6         int n =5;
7         for(int i =1; i <= n; i++){
8             for (int j = 1; j <= n; j++) {
9                 System.out.print(j);
10            }
11            System.out.println();
12        }
13    }
14 }
15
16
17
18
19

```

Run First

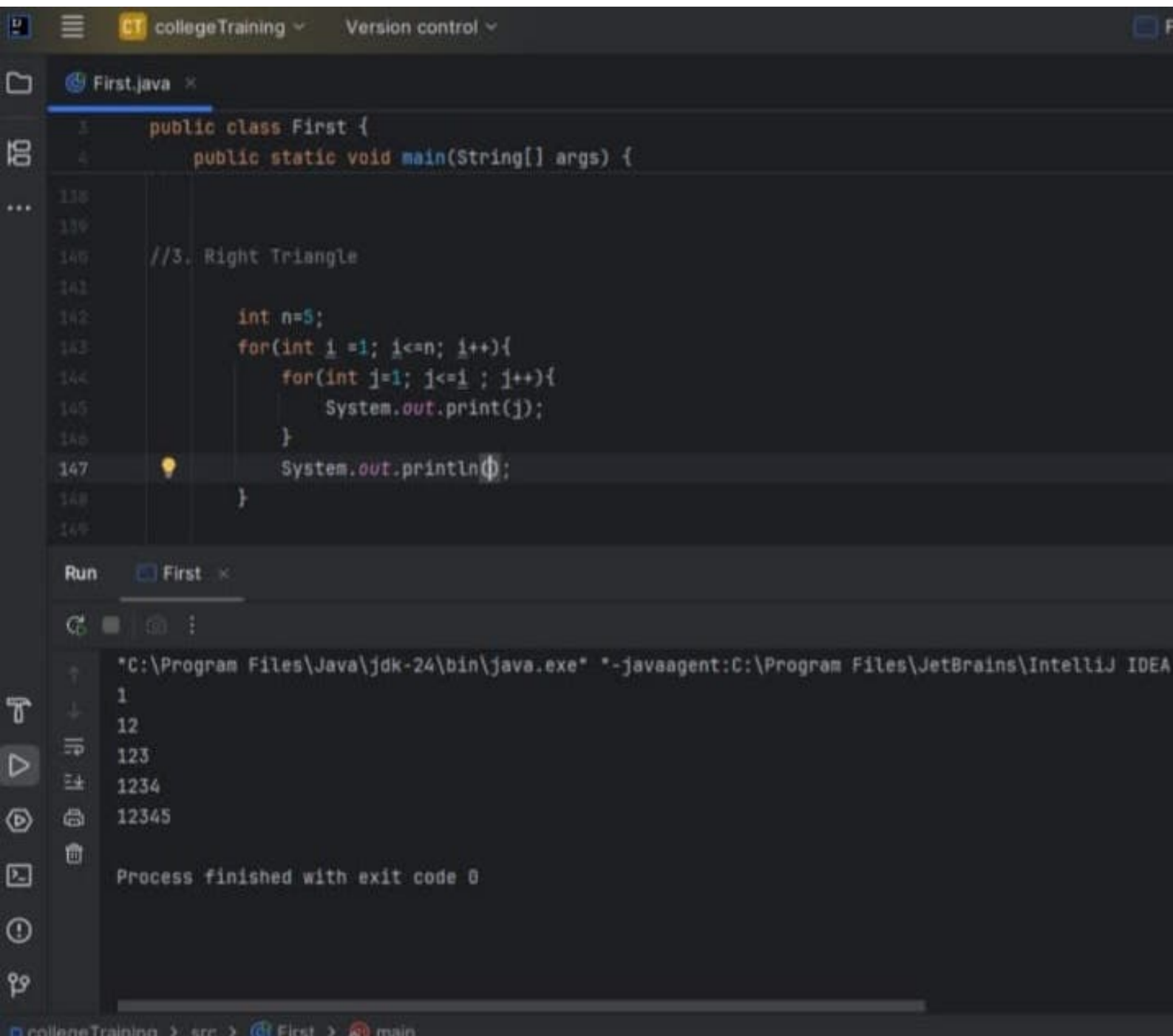
Run

```

*C:\Program Files\Java\jdk-24\bin\java.exe* *-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA
12345
12345
12345
12345
12345

Process finished with exit code 0

```



4. Inverted diamond

logic: • $n=5$ and start from 3

- Outer loop $i = 1$ to n
- Count of number for row $= n - i + 1$ (decreasing)
- Value per row $= \text{start} + i - 1$
- Left spaces $= i - 1$ (incr. each row for Centre alignment)
- print spaces first, then value (count) times with space between each.

Code:

```
int n = 5, start = 3;
for (int i = 1; i <= n; i++) {
    int count = n - i + 1;
    int value = start + i - 1;
    for (int j = 0; j < i - 1; j++) {
        system.out.print(" ");
    }
    for (int j = 0; j < count; j++) {
        system.out.print(value);
        if (j < count - 1) {
            system.out.print(" ");
        }
    }
    system.out.println();
}
```

```
3 3 3 3 3
 4 4 4 4
   5 5 5
    6 6
     7
```

```

3      public class First {
4          public static void main(String[] args) {

150      // 4. Inverted Diamond
151          int n=5;
152          int start = 3;
153          for(int i =1; i<=n; i++){
154              int count = n-i+1;
155              int value = start + i -1;
156              for (int j = 0; j < i-1; j++) {
157                  System.out.print(" ");
158              }
159              for (int j = 0; j < count; j++) {
160                  System.out.print(value);
161                  if( j < count-1){
162                      System.out.print(" ");
163                  }
164              }
165              System.out.println();
166          }
    
```

```

+ *C:\Program Files\Java\jdk-24\bin\java.exe* *-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA
+ 3 3 3 3 3
+ 4 4 4 4
+ 5 5 5
+ 6 6
+ 7
    
```

5. Sandwich pattern

logic: • set $n = 5$, $start = 3$

- outer loop $i = 1$ to n
- $m = \min(i-1, n-i) \rightarrow$ distance from nearest edge
- $value = start + m$ (goes $3 \rightarrow 4 \rightarrow 5 \rightarrow 4 \rightarrow 3$)
- $count = n - 2 * m$ (for centre alignment)
- print spaces then value (count) times with space between each other.

code: `int n = 5, start = 3;`

```
for (int i = 1; i <= n; i++) {  
    int m = Math.min(i-1, n-i);
```

```
    int value = start + m;
```

```
    int count = n - 2 * m;
```

```
    for (int j = 0; j < 2 * m; j++) {
```

```
        system.out.print(" ");
```

```
    }
```

```
    for (int j = 0; j < count; j++) {
```

```
        system.out.print(value);
```

```
        if (j < count - 1) {
```

```
            system.out.print(" ");
```

```
        }
```

```
    }
```

```
    system.out.print("\n");
```

```
}
```

```
3 3 3 3 3  
 4 4 4  
   5  
  4 4 4  
3 3 3 3 3
```

First.java

×

3

public class First {

4

public static void main(String[] args) {

169

//5. Sandwich Pattern

170

int n = 5, start= 3;

171

for (int i = 1; i <= n; i++) {

172

int m = Math.min(i-1, n-i);

173

int value = start + m;

174

int count = n-2*m;

175

for (int j = 0; j < 2*m; j++) {

176

System.out.print(" ");

177

}

178

for (int j = 0; j < count; j++) {

179

System.out.print(value);

180

if(j < count -1){

181

System.out.print(" ");

182

}

183

}

184

System.out.println();

185

}

186

Run

First

×

↑

↓

⇌

⌕

3 3 3 3 3

4 4 4

5

4 4 4

3 3 3 3 3

collegeTraining

>

src

>

First

>

main

6. Incrementing diamond Pattern

Logic : • $n = 5$, $start = 3$

- 2 loops, upper ($i = 1$ to n) & lower ($i = n - 1$ to 1)
- Row i first value ($start + i - 1$) exactly i times with spaces.
- Value increases $3 \rightarrow 7$ then decreases $6 \rightarrow 3$ count mirrors

```
int n = 5, start = 3;
for (int i = 1; i <= n; i++) {
    for (int j = 0; j < i; j++) {
        system.out.print(start + i - 1);
    }
    system.out.println();
}
for (int i = n - 1; i >= 1; i--) {
    for (int j = 0; j < i; j++) {
        system.out.print(start + i - 1);
    }
    system.out.println();
}
```

```
3
4 4
5 5 5
6 6 6 6
7 7 7 7 7
6 6 6 6
5 5 5
4 4
3
```


First.java

```

3      public class First {
4          public static void main(String[] args) {
187              //6. Incrementation Diamond Pattern
188              int n =5,start =3;
189              for(int i =1; i<= n; i++){
190                  for(int j =0; j< i; j++){
191                      System.out.print(start + i -1);
192                  }
193                  System.out.println();
194              }
195              for (int i = n-1; i >= 1 ; i--) {
196                  for (int j = 0; j < i; j++) {
197                      System.out.print(start + i-1);
198                  }
199                  System.out.println();
200              }
201          }

```

Run
First

```

3
44
555
6666
77777
6666
555
44
3

```